

Appendix

CoP Survey - Published

Start of Block: Opening of the survey

Q1.1

Welcome to the MARCO-BOLO Survey

This survey is part of the MARCO-BOLO (MARine Coastal BiODiversity Long-term Observations) project **strengthening biodiversity observations for decision-making** in and beyond the EU and its member states.

We want to **understand the data needs** of different stakeholders working in biodiversity conservation and sustainable use. Understanding your data-related needs will help us create better data products to support your day-to-day work.

The survey **targets the broader biodiversity community**. We encourage biodiversity data users and non-users to share their experiences with us. It includes up to 25 questions and will take **10-12 minutes** to be completed.

For general questions or comments regarding the study, please contact:

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alice.vadrot@univie.ac.at

If you encounter technical difficulties, please contact:

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For further information on the EU project MARCO-BOLO please visit our website:
<https://marcobolo-project.eu/>

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Q1.2 Funded by the European Union under the Horizon Europe Programme, Grant Agreement No. 101082021 (MARCO-BOLO). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Research Executive Agency (REA). Neither the European Union nor the granting authority can be held responsible for them.

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Q1.3 **Data protection**

Your responses to MARCO-BOLO survey questions will be recorded and stored on secure servers. Recorded data will not include any personal identification; hence it will not be possible to identify you as the source of information. Data will be processed for scientific purposes during the phase of data analysis. Data will be shown — in project reports, scientific journal publications, conference presentations or other project-related outputs — in anonymised form only. Results may be used in further studies. None of the provided personal data will be passed on to third parties.

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Q1.4 **Informed consent**

Your participation in this study is voluntary. You can choose not to participate or to leave the survey at any point. However, once you have submitted your answers, there will be no way to withdraw them because the survey does not contain any identifying information.

By participating in this online survey, you confirm that:

- You are 18 years or older and competent to provide consent
- You have read the above information about this survey and the informed consent procedure
- You have been fully informed about the aims and purposes of the MARCO-BOLO Project
- You have been given the opportunity to ask questions regarding the purpose of the study
- You understand that your participation is fully anonymous and that no personal details about

you will be recorded

- You agree that your responses, collected in the survey, will be used for scientific purposes and confirm that you have no objection to this data being published in project-related publications in a way that does not reveal your identity
- Information may be shared between any of the other researcher(s) and partners participating in this project in anonymised form. All information provided by you will be treated as confidential.

This consent form is made pursuant to the relevant national, European and international data protection laws and regulations and personal data treatment obligations. More specifically, it complies with the EU General Data Protection Regulation (2016/679) on the protection of natural persons with regard to the processing of personal data and on the free movement of such data.

Q1.4

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Q1.5 I have read the information and consent form for participation in the study and I agree to participate in the survey.

- ☐ I agree
- ☐ I disagree

End of Block: Opening of the survey

Start of Block: Demographics

Q2.1 How old are you?

- ☐ 18-25
- ☐ 25-40
- ☐ 41-55
- ☐ 56-70
- ☐ 71+

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Q2.2

In which country is your institution based?

- ☐ Afghanistan
- ☐ Albania
- ☐ Algeria
- ☐ Andorra

- Angola
- Antigua and Barbuda
- Argentina
- Armenia
- Australia
- Austria
- Azerbaijan
- Bahamas
- Bahrain
- Bangladesh
- Barbados
- Belarus
- Belgium
- Belize
- Benin
- Bhutan
- Bolivia
- Bosnia and Herzegovina
- Botswana
- Brazil
- Brunei Darussalam
- Bulgaria
- Burkina Faso
- Burundi
- Cambodia
- Cameroon
- Canada
- Cape Verde
- Central African Republic
- Chad
- Chile
- China

- Colombia
- Comoros
- Congo, Republic of the...
- Costa Rica
- Côte d'Ivoire
- Croatia
- Cuba
- Cyprus
- Czech Republic
- Democratic Republic of the Congo
- Denmark
- Djibouti
- Dominica
- Dominican Republic
- Ecuador
- Egypt
- El Salvador
- Equatorial Guinea
- Eritrea
- Estonia
- Ethiopia
- Fiji
- Finland
- France
- Gabon
- Gambia
- Georgia
- Germany
- Ghana
- Greece
- Grenada
- Guatemala

- Guinea
- Guinea-Bissau
- Guyana
- Haiti
- Honduras
- Hong Kong (S.A.R.)
- Hungary
- Iceland
- India
- Indonesia
- Iran
- Iraq
- Ireland
- Israel
- Italy
- Jamaica
- Japan
- Jordan
- Kazakhstan
- Kenya
- Kiribati
- Kuwait
- Kyrgyzstan
- Lao People's Democratic Republic
- Latvia
- Lebanon
- Lesotho
- Liberia
- Libyan Arab Jamahiriya
- Liechtenstein
- Lithuania
- Luxembourg

- Madagascar
- Malawi
- Malaysia
- Maldives
- Mali
- Malta
- Marshall Islands
- Mauritania
- Mauritius
- Mexico
- Micronesia, Federated States of...
- Monaco
- Mongolia
- Montenegro
- Morocco
- Mozambique
- Myanmar
- Namibia
- Nauru
- Nepal
- Netherlands
- New Zealand
- Nicaragua
- Niger
- Nigeria
- North Korea
- Norway
- Oman
- Pakistan
- Palau
- Panama
- Papua New Guinea

- Paraguay
- Peru
- Philippines
- Poland
- Portugal
- Qatar
- Republic of Moldova
- Romania
- Russian Federation
- Rwanda
- Saint Kitts and Nevis
- Saint Lucia
- Saint Vincent and the Grenadines
- Samoa
- San Marino
- Sao Tome and Principe
- Saudi Arabia
- Senegal
- Serbia
- Seychelles
- Sierra Leone
- Singapore
- Slovakia
- Slovenia
- Solomon Islands
- Somalia
- South Africa
- South Korea
- Spain
- Sri Lanka
- Sudan
- Suriname

- Swaziland
- Sweden
- Switzerland
- Syrian Arab Republic
- Tajikistan
- Thailand
- The former Yugoslav Republic of Macedonia
- Timor-Leste
- Togo
- Tonga
- Trinidad and Tobago
- Tunisia
- Turkey
- Turkmenistan
- Tuvalu
- Uganda
- Ukraine
- United Arab Emirates
- United Kingdom of Great Britain and Northern Ireland
- United Republic of Tanzania
- United States of America
- Uruguay
- Uzbekistan
- Vanuatu
- Venezuela, Bolivarian Republic of...
- Viet Nam
- Yemen
- Zambia
- Zimbabwe

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Q2.3 What type of institution are you mainly affiliated with?

- ☐ National governmental institution
- ☐ Local or regional governmental institution
- ☐ European Union and its agencies
- ☐ International organization
- ☐ Nongovernmental organization (NGO)
- ☐ University
- ☐ Other academic research institution
- ☐ Business and Industry
- ☐ Local association
- ☐ Other _____

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Q48 To which of these scientific organisational units would you belong to? (Multiple answers possible)

- ☐ Natural sciences
- ☐ Data science and informatics
- ☐ Social sciences
- ☐ Business and economy
- ☐ Law
- ☐ Other _____

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Q43 What department / sector are you working in?

- Science, education and research
- Environment and conservation
- Agriculture and / or fisheries
- Energy and / or transport
- Other _____

Page Break

Q2.4 What is your main area of work?

- Law, policy and decision-making
- Academia, research and education
- Data management
- Conservation and environmental management
- Management and administration
- Media and communication
- Public engagement
- Product development and construction
- Other _____

Page Break

Q2.5 What ecosystem do you engage with? (Multiple responses possible)

- Marine
- Coastal

- Freshwater
- Terrestrial
- None of these

End of Block: Demographics

Start of Block: Monitoring

Q3.1

In this part of the survey, we are particularly interested in your experiences with **biodiversity monitoring**.

Monitoring is broadly defined as;

"The repeated sampling and analysis in time or space of one or more ecosystem components."
-Zampoukas et al. (2014)

Q3.2 How well are biodiversity monitoring services established in your country?

- Terrible
- Poor
- Average
- Good
- Excellent
- Don't know

Page Break

Q3.3 To what extent do the following factors determine the quality of biodiversity monitoring?

	Not at all important	Slightly important	Moderately important	Very important	Extremely important	Don't know
Coverage (of species and ecosystems)	•	•	•	•	•	•
Long-term funding	•	•	•	•	•	•
Data and methods sharing	•	•	•	•	•	•
Technical infrastructure	•	•	•	•	•	•
Improvement of metadata	•	•	•	•	•	•
Usability	•	•	•	•	•	•
Political support	•	•	•	•	•	•
Private investment	•	•	•	•	•	•
Data curation	•	•	•	•	•	•
Interoperability (the ability of monitoring systems to connect in a coordinated way)	•	•	•	•	•	•
Stakeholder involvement	•	•	•	•	•	•
Citizen science	•	•	•	•	•	•
Other	•	•	•	•	•	•

Q3.4 Have you participated in biodiversity monitoring activities?

- ☐ Yes
- ☐ No

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Q3.5 What could improve biodiversity monitoring?

Please rank the following solutions via **drag and drop** from top (very urgent) to bottom (less urgent):

- _____ Standardized protocols or templates
- _____ Aligning reporting requirements with specific indicators
- _____ Increase data standardization
- _____ New technologies to integrate different methods and data
- _____ Additional, consistent, and long-term financial resources
- _____ More and better coordination across countries, institutions, and sectors
- _____ Expand and adapt capacity building and training
- _____ Embracing citizen science initiatives
- _____ Expanding reporting obligations
- _____ More data points and better coverage

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Q3.6

What other solutions do you see?

End of Block: Monitoring

Start of Block: Data use

Q4.1 In the following we would like to know more about your experiences of working with biodiversity **data**.

Q4.2 What applies to you? (Multiple responses possible)

- I produce biodiversity data
- I manage biodiversity data
- I use biodiversity data
- I do NOT use or produce biodiversity data

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AB HIER NON DATA USER

Q5.1 What prevents you from using biodiversity data in your work? (Multiple responses possible)

- My job does not need it
- It is too difficult to understand
- It is too difficult to apply
- The data I would need is not supplied
- I do not know where to find it
- Data not available in specific language
- I have to pay for data
- Other _____

Page Break

Q5.2 Does your institution encourage the use of biodiversity data?

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

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Q44 For what purpose would you need biodiversity data? (Multiple answers possible)

- Scientific research
- Education
- Communication
- Reporting
- Policy-making
- Decision-making
- Spatial planning
- Environmental impact assessment
- Conservation measures
- Product development
- Management and accounting
- Protected area management
- Indicator development
- Policy evaluation
- Other: _____

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Q6.5 What type of biodiversity-related data would you need? (Multiple answers possible)

- Physical data
- Geological data
- Chemical data
- Biological data
- Socio-economic data
- Numeric data
- Visual data
- Acoustic data
- Genetic data
- Fishery data
- Satellite data
- Pollution data
- Other _____
- None of these

DATA NON USER ENDE

Page Break

AB HIER DATA USER/PRODUCER/MANAGER

Q5.3 Which of the following factors would enhance the use of biodiversity data in your work?
(Multiple answers possible)

- Institutional obligation
- Visualization of data
- Opportunities to increase your data handling skills

- Easier access to data
- Better cooperation with other data users
- Validation and certification of data
- Information available in specific language
- Other _____
- Nothing

Q6.1 For what purpose do you use biodiversity data? (Multiple answers possible)

- Scientific research
- Education
- Communication
- Reporting
- Policy-making
- Decision-making
- Spatial planning
- Environmental impact assessment
- Conservation measures
- Product development
- Management and accounting
- Protected area management
- Indicator development
- Policy evaluation
- Other: _____

Page Break

Q6.2 Does your institution encourage the use of biodiversity data?

- Strongly disagree
- Somewhat disagree
- Neither agree nor disagree
- Somewhat agree
- Strongly agree

Page Break

Q6.3 How much time do you spend on dealing with **data in general**?

- Below 10%
- 10% to 30%
- 30% to 50%
- 50% to 70%
- Most of my time

Page Break

Q6.4 How much time do you spend specifically with **biodiversity data** related tasks?

- Below 10%
- 10% to 30%
- 30% to 50%
- 50% to 70%
- Most of my time

Page Break

Q45 What type of biodiversity-related data do you use? (Multiple answers possible)

- Physical data
- Geological data
- Chemical data
- Biological data
- Socio-economic data
- Numeric data
- Visual data
- Acoustic data
- Genetic data
- Fishery data
- Satellite data
- Pollution data
- Other _____
- None of these

Page Break

Q6.6 What data **products and tools** do you need most urgently? (Multiple responses possible)

- Maps
- Graphs
- Scenarios
- Models
- Tools to integrate data
- Other _____

Page Break

Q6.7 What **challenges** do you experience when using biodiversity data?

	Never	Sometimes	About half the time	Most of the time	Always	Don't know
Data is not accessible to me	☐	☐	☐	☐	☐	☐
I don't know how to get hold of the data	☐	☐	☐	☐	☐	☐
I don't know the data I need exists	☐	☐	☐	☐	☐	☐
I don't know, what data I need	☐	☐	☐	☐	☐	☐
Data is not useable/interoperable with my work/project	☐	☐	☐	☐	☐	☐
Lack of metadata	☐	☐	☐	☐	☐	☐
Data interpretation is difficult without expert's input	☐	☐	☐	☐	☐	☐
Lack of consistency	☐	☐	☐	☐	☐	☐
There are issues in technical maintenance (Website errors, etc)	☐	☐	☐	☐	☐	☐
The data is not delivered at the right time	☐	☐	☐	☐	☐	☐
Data is too expensive	☐	☐	☐	☐	☐	☐
Other	☐	☐	☐	☐	☐	☐

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Q6.8 **Essential Variables** are priority variables for monitoring. They aim to standardize monitoring data to a certain degree to make data interoperable.

Q6.9

To what extent do you use **Essential Variables** in your work?

	Never	Once	Regularly	Daily
EBV (Essential Biodiversity Variables)	☐	☐	☐	☐
EOV (Essential Ocean Variables)	☐	☐	☐	☐
Other	☐	☐	☐	☐

End of Block: Data use

Start of Block: Data repositories

Q7.1 Several global and regional **data repositories** have been created to store and facilitate the use of biodiversity data.

Please answer the following questions to indicate your familiarity with and use of these repositories.

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Q7.3

How often do you use the following data repositories?

	Never	Once	Regularly	Daily
EMODnet (European Marine Observation and Data Network)	☐	☐	☐	☐

OBIS (Ocean Biodiversity Information System)	☐	☐	☐	☐
GBIF (Global Biodiversity Information Facility)	☐	☐	☐	☐
National repositories	☐	☐	☐	☐
Project based repositories	☐	☐	☐	☐
Other	☐	☐	☐	☐

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Q41 Is there a repository that you can recommend?

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Q7.4 What factors would increase your use of data from data repositories? (Multiple responses possible)

- ☐ Better accessibility
- ☐ Better interoperability
- ☐ Better display of metadata
- ☐ Transparent quality control
- ☐ Language options for the website
- ☐ Easy technical access

- Guidance on how to use data repositories
- More effort to advertise and promote available data repositories
- Other _____
- Nothing

Page Break

Q46 Do you feed your data into data repositories?

- Never
- Sometimes
- About half the time
- Most of the time
- Always

Page Break

Q6.10 On a scale from 1 to 5, how would you rate your data literacy?

- Extremely bad
- Somewhat bad
- Neither good nor bad
- Somewhat good
- Extremely good

End of Block: Data repositories

Start of Block: Registration

Q8.1 REMOVED FOR ANONYMIZATION is creating a Community of Practice to bring biodiversity data generators and data users (REMOVED FOR ANONYMIZATION CoP) together to co-design and co-create biodiversity tools and services that are fit for purpose and meet the needs of users at local, national, regional and international levels.

Are you interested in joining the REMOVED FOR ANONYMIZATION Community of Practice?

- ☐ Yes
- ☐ No

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Q8.2 What level of engagement are you interested in?

- ☐ Very high level - Participate in Co-design Workshops
- ☐ High level - Participate in CoP Events

Medium level - Review expected REMOVED FOR ANONYMIZATION outputs/products

Low level - Sign up to an email distribution list to receive REMOVED FOR ANONYMIZATION project updates

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Q8.3 Your e-mail address:

Page Break

Q42 Please provide your details.

- Title _____
- First name _____
- Last name _____
- Professional title _____
- Name of Institution _____
- E-mail _____

End of Block: Registration